PredictEvent

Event-Driven Congestion Intelligence for Bengaluru Traffic Police

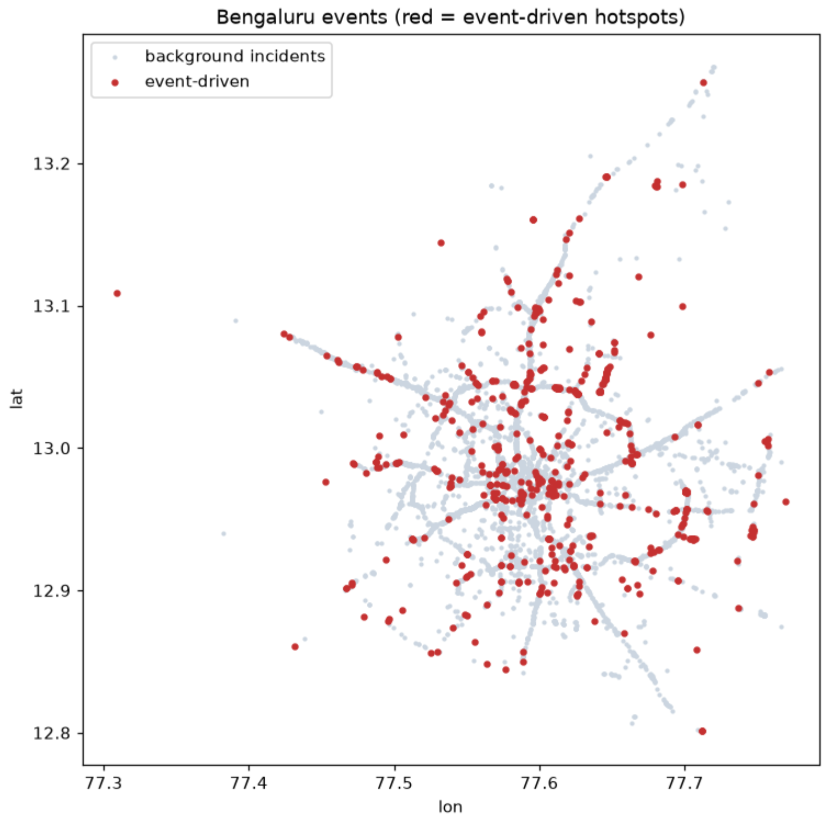
Theme 2 | BTP × Flipkart Hackathon | Forecast impact → recommend manpower, barricading & diversion

The Problem

- Rallies, festivals, sports, processions, VIP movement, construction & sudden incidents choke Bengaluru's corridors.
- Event impact is not quantified in advance — which corridor, how hard, tomorrow?
- Resource deployment is experience-driven — manpower & barricades decided by gut feel.
- No post-event learning loop — the same mistakes repeat.
- Goal: turn 8,173 historical events into a forward-looking deployment planner.

Our Solution

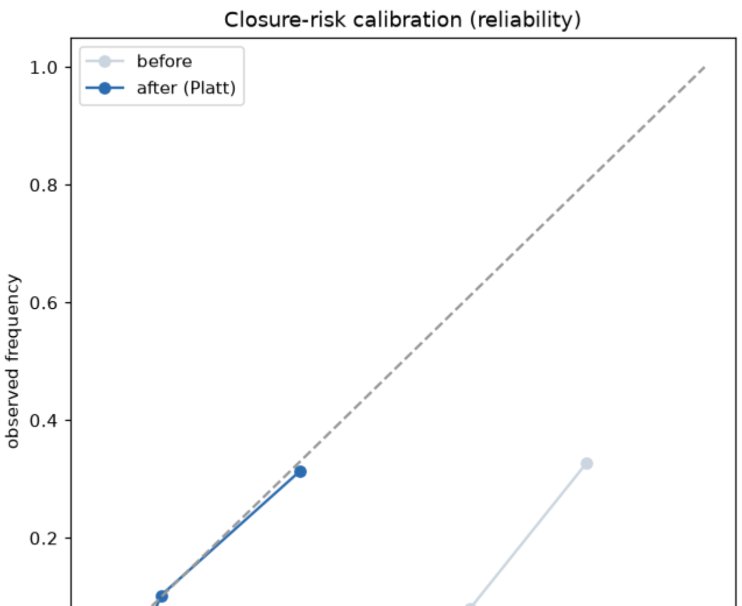
- **A laptop-deployable decision-support engine (no GPU) with 3 part**
  - – Hotspot Load Forecaster → forecast load per corridor for tomorrow
  - – Closure-Risk Classifier → barricade & RED/AMBER/GREEN triage
  - – OR Optimization Engine → SLA-guaranteed officers + coverage-optim
- **Uses ONLY the provided ASTRAM dataset. Interpretable, deployab**



ML Results — honest, time-based split

Component	Result	Baseline
Hotspot Load Forecaster	R^2 0.536 · MAE 2.19	seasonal-naive R^2 −0.32
Closure-Risk Classifier	ROC-AUC 0.81 · recall 0.83	random PR-AUC 0.072
Duration Predictor	not reliably learnable	(data limitation, see findings)

Operating point: F2-max on forward-chained OOF preds — catches 83% of closures (a missed closure costs BTP more).
Top drivers (permutation importance): event_cause ≫ corridor > location > vehicle type.
Deep ST-model: MAE 2.12 vs GBM 2.24 (~5% better, R^2 tied); ensemble R^2 0.55. GBM kept in prod.
Calibrated (isotonic): ECE 0.33→0.03; + safety floor max(model, historical) so VIP (rare) isn't under-flagged.
Cross-validated (5-fold expanding): hotspot R^2 0.56±0.14, closure AUC 0.73±0.05 — spread, not cherry-pick.



OR Optimization Engine — the differentiator

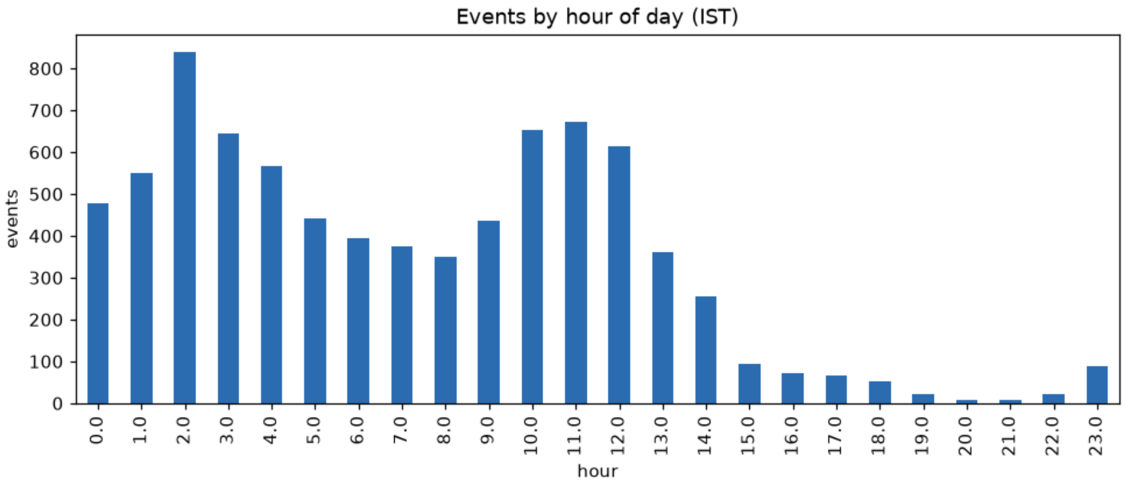
- We replaced heuristic rules with the forecast→optimize pattern used in deployed police systems.
- Manpower = M/M/c queueing (Erlang-C): officer count sized so expected response wait ≤ SLA (e.g. 5 min) — a number WITH a service-level guarantee.
- Barricade placement = Maximum Coverage Location Problem (MCLP): place P units to cover max risk×load-weighted demand (PuLP/CBC).
- Risk×load weighting: both high-traffic AND high-risk corridors, per OR location-allocation literature.
- Bengaluru real-world context (researched): corridor congestion (ORR East 1.86 .. Magadi 1.13), commute peaks, monsoon, festivals, metro works, heavy-vehicle bans — all scale the deployment.
- Every number is traceable — no black box. Evidence: SMU IJCAI-2019, Vlahogianni 2019, Miao/Easa 2021.

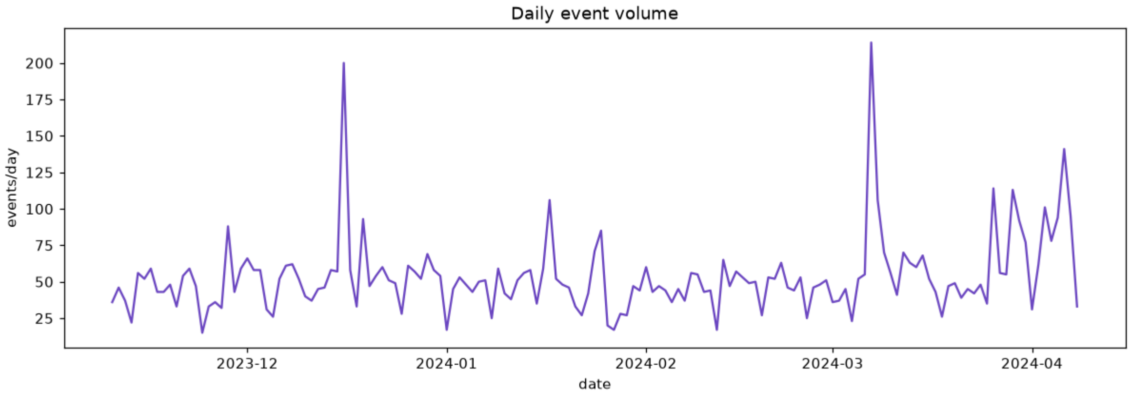
Data Findings That Build Trust

- **Timestamps are local IST mislabeled '+00' — proven via bimodal rush-hour signature; we use stored hour as-is.**
- **closed_datetime is administrative ticket-close, NOT on-ground clearance → exact duration not learnable. We refuse to over-claim it.**
- **We caught & removed target leakage: end-coordinate flag is filled BECAUSE a closure happened (98% vs 0%) — faked ROC-AUC 1.0.**
    - – Permutation importance exposed it; dropping it gives the honest 0.81.
- **Event-driven causes (public_event, procession, VIP, protest) have 40–80% closure rates — the true Theme-2 targets.**

Prototype — Live Demo

- **Event Simulator — type an event (e.g. IPL match @ Chinnaswamy) → instant risk %, tier, personnel, barricade & diversion.**
- **Hotspot Forecast + Manpower — pick a day → corridor load + recommended personnel.**
- **Live Event Triage — every event ranked by closure-risk with action tier + map.**
- **Analytics — historical patterns + feature importance.**





Where It Fits in BTP's Stack

- BATCS optimizes the SIGNALS — adaptive timing at 165 junctions (CoSiCoSt). Not people.
- ASTRAM SEES the city — awareness, 15-min alerts, microsimulation. (Our dataset's source.)
- Neither produces a prescriptive police-deployment plan.
- PredictEvent DECIDES the deployment — turns ASTRAM's event data into an officer count with a service-level guarantee + coverage-optimal placement.
- A prescriptive module on top of the stack BTP already runs — it extends ASTRAM, doesn't compete.

Why It Wins · Roadmap

- **Solves BTP's actual #1 pain — event manpower planning, not a tech demo.**
- **OR-optimized, SLA-guaranteed deployment — not heuristic rules.**
- **Honest, leakage-checked metrics — survives a technical jury.**
- **Plugs into ASTRAM; deployable today; provided-data only.**
- **Roadmap:**
  - – Live ASTRAM API feed → real clearance timestamps
  - – Capacitated city-wide MCLP + IP shift-scheduling (ILS/tabu fast replanning)
  - – Mobile officer app + automatic retraining loop